

Examiner reports

Q1.

Most candidates replaced $\frac{8}{x^4}$ by $8x^{-4}$ and differentiated correctly to score full marks for part (a).

Some other candidates failed to score the accuracy mark because they included a '+c' in their

answer. A very large majority of candidates realised that the value of $\frac{dy}{dx}$ at $x = 1$ was required in part (b), but a small minority did not appreciate that the value of y was also required. Although many candidates found the equation of the tangent correctly, it was not uncommon to see others finding the equation of the normal instead.

In part (c), a majority of candidates equated their expression for $\frac{dy}{dx}$ to zero and attempted to solve the resulting equation, although only about half of the candidates obtained the correct coordinates of M . Those who had wrong powers of x in their answer to part (a) were restricted to just two marks in part (c). The most common incorrect methods seen either involved solving

$\frac{d^2y}{dx^2} > 0$ or looking for a link between the tangent in part (b) and the stationary point.

In part (d)(i), a high percentage of candidates were able to integrate $\frac{8}{x^4}$ correctly, but errors in the integration of $(x + 3)$ or the omission of the constant of integration resulted in a significant proportion of these candidates failing to score the final mark. Since answers to part (d)(ii) required explicit evidence that the answer to part (d)(i) had been used; the few candidates who just wrote down the answer without any other working gained no credit. However, most candidates applied the 'Hence', showing the values 2 and 1 substituted into their answer to part (d)(i) and carrying out the relevant subtraction. However, a significant number of candidates made arithmetical errors in carrying out the subtraction and so did not score the accuracy mark.

Only the more able candidates appreciated what was required in the final part of the question. The common wrong answers included 0 (presumably from the x -axis being a tangent), 12 (the y value from part (b)) and 5.5 (forgetting to consider the direction of the translation).

Q2.

Part (a) was well answered by the majority of candidates with many fully correct responses seen. Most candidates used substitution of $u = (x^3 - 1)$ to give $y = u^6$. Although $(x^3 - 1)^5$ was usually seen, errors did occur with the associated function of x .

Part (b)(i) was very well answered, with a majority of candidates earning both marks. The major error was from candidates who did not treat this as a product and simply differentiated x to get 1

and $\ln x$ to get $\frac{1}{x}$.

Part (b)(ii) Again, this part was very well answered. Many fully correct responses were seen. The main error was with candidates who wrote the gradient as $1 + \ln e$ which earned the method mark. Then this was often evaluated as $1 + \ln e = 1$ or left unsimplified. Several candidates who

obtained $\frac{dy}{dx} = 2$ then went on to use a gradient of $-\frac{1}{2}$.

Q3.

This question was also done well, with again nearly half the entry gaining full marks.

In part (a)(i) most candidates found the derivatives $\frac{dx}{dt}$ and $\frac{dy}{dt}$ successfully and went on to

divide to find $\frac{dy}{dx}$ correctly. The most common, although rare, error was the omission of the

minus sign when differentiating e^{-2t} . Occasionally a candidate used $\frac{dx}{dt} / \frac{dy}{dt}$.

In (a)(ii) almost all candidates gained this mark for a correct tangent equation using their numerical gradient from part (a)(i). Only a few took the negative reciprocal to find the equation of the normal.

In part (b) most candidates were successful too. Whilst some used the quickest method of substituting $e^t = \frac{x}{3}$ or $e^{2t} = \frac{x^2}{9}$, others used logarithms. Candidates were required to conclude with the correct equation, which some failed to do; a verification was given no credit.

Q4.

In part (a) finding $\frac{dV}{dt}$ caused few problems apart from those who chose to multiply by 4 before

differentiating. Some will insist on adding $+c$ to their derivative despite advice given in previous reports.

In (b)(i) the concept of “rate of change” seemed to be better understood this time, helped by there being only one derivative to choose from; careless arithmetic when trying to simplify

$0.75 - 3$ spoiled many solutions.

Part (b)(ii) Most candidates realised that the volume was decreasing when $t = 1$ but many

failed to give an adequate reason; most neglected to state the crucial fact that the rate of change was negative.

In part (c)(i) candidates were expected to solve an equation of the form $\frac{3}{4}t^2 - 3 = 0$ but most

chose to guess that $t = 2$ was the solution and attempted to verify this fact.

Part (c)(ii) Once again the explanation as to why the stationary value was a minimum value was inadequate, very often taking no account of the value of t found in the previous part of the

question. Some candidates who correctly found that the second derivative was $\frac{3t}{2}$ and

proceeded to solve the equation $\frac{3t}{2} = 0$ often obtained a value such as 6 and stated that this positive solution indicated a minimum value.

Q5.

Many candidates did not have enough room to complete this question in the space available, often because of over-elaborating their answer to part (b)(ii).

Part (a) was answered correctly in the majority of cases.

Part (b)(i) was also done well, with only a small number making slips with signs, or square-rooting when they should have squared.

Part (b)(ii) had only 1 mark allocated and a request to “write down” the required equation. Although approximately 25% of the candidates did write down the correct equation, $x = 4$, of the normal at the maximum point, a far more popular but wrong answer was $y = 8$. Many other candidates produced half a page of working to reach an equation of a line which was neither vertical nor horizontal.

It was pleasing to see a much better response in part (c)(i) with many more finding the correct equation of the normal at P , although many failed to reduce it to a form with positive integers a, b

and, in particular, c . It was not uncommon to see the final answer left as ‘ $2x + 3y = \frac{99}{4}$ ’, just one

step away from the correct answer ‘ $8x + 12y = 99$ ’. In general, only the more able candidates, who made good use of the given diagram to show that the normal at M was vertical, gained credit for their answers to part (c)(ii) of the question.

Q6.

In part (a)(i) the majority of candidates were able to differentiate e^{2x} and apply the product rule correctly but a few made an error in one or other of these processes. Occasionally an attempt to simplify was incorrect.

In (a)(ii) when finding the equation of the tangent (a line) it is essential to find the gradient (a constant) at the requisite point first. Most did this but it is disconcerting, at this level, to find candidates who give a non-linear equation as their answer by failing to do the first

step. A few added $e^2 + 2e^2$ wrongly or lost exactness by evaluating $3e^2$, and a few found the equation of the normal.

In (b) although the quotient rule was known by almost everyone, and almost all earned the first mark, some appalling algebra and trigonometry abounded thereafter. A few failed even to get the first mark as they put a term in the wrong place on the numerator, had $\sin x$ or $\cos x$ instead of $\sin 3x$ or $\cos 3x$ in one of their differentials or else their denominator was $1 + \cos^2 3x$ instead of

$(1 + \cos 3x)^2$. A few failed to use the chain rule when differentiating $\sin 3x$ or $\cos 3x$. Those who missed the brackets around $(1 + \cos 3x)$ in the numerator sometimes recovered but far too many falsely cancelled $\cos 3x$ at this stage. If the numerator was correctly expanded another mark was available, although $\cos 3x^2$ and $\cos^3 3x$ instead of $\cos^2 3x$ for $(\cos 3x)^2$ was common; this was recoverable, but $\cos 9x^2$ was not. After that more incorrect cancellation was seen. Taking out a common factor of 6 was the safest route forward, as those who tried to deal with $6\cos^2 3x + 6\sin^2 3x$ often put it equal to 1 or 18 instead of 6. Even being correct to that stage did not ensure completion as, again there was false cancellation. Those who split the terms, then used $(1 - \cos^2 3x)$ for $\sin^2 3x$ and factorised to $(1 - \cos 3x)(1 + \cos 3x)$ generally completed successfully. It was quite extraordinary how many methods with completely wrong working

landed up at $\frac{6}{1 + \cos 3x}$.

Q7.

Most candidates scored the marks for the two derivatives with relatively few omitting the minus signs. Some candidates managed to drop the 2 from $\cos 2\theta$. Many candidates brought the

commonly used parameter t into their expression using $\frac{dx}{dt} = -\sin \theta$ or similar, but were not penalised. Virtually all knew how to use the chain rule, but many got mixed up with the coefficients in both expanding $\sin 2\theta$ and simplifying the expression. The common error was to find $k = 3$. Some candidates got a result in terms of $\sin \theta$ and some got no result at all, but any value of k , however found or invented, was allowed in part (a)(ii).

Part (a)(ii) was generally done well with most candidates demonstrating they both knew and could use the relation between the gradients of the tan gent and the normal, although a few

found an equation of the tan gent. The evaluation using $\theta = \frac{\pi}{3}$ was generally accurately done; however candidates had to have derived $k = 6$ correctly to gain the final mark for the equation of the normal.

In part (b) it was expected here that candidates would note the whole question was about the use of double angle formulae, and many did. Most got as far as at least trying to express $\sin^2 x$ in term of $\cos 2x$ and of those, many did it correctly, even if sometimes there was a mix of variables. The integration of $\cos 2x$ was usually accurate, with the common error being to double $\sin 2x$ rather than halve it. There were few sign errors. Virtually all candidates who achieved an

integral knew how to use the limits, but the real test here was in handling $\sin 2\left(-\frac{\pi}{4}\right)$; many candidates lost credit through not showing their evaluation or getting the sign wrong.

However, many candidates, about 25% of the entry, went on to obtain the correct answer and give it in an exact form, although some did make a mistake with the signs, particularly the three minus signs involved with the lower limit. Of the few candidates who attempted the integral using parts most made little progress beyond the first line of working, not realising that in order to proceed with the integral they needed to change $\cos^2 x$ back to $\sin^2 x$, and manipulate the resulting expression.

Q8.

In part (a) about a third of the candidates could not evaluate the expression in e correctly, the

term $e^2\left(\frac{1}{e}\right)^2$ causing many problems.

In part (b) most candidates scored the marks for the implicit differentiation, with many using the product rule correctly. The common errors were to keep the y in the derivative, of $2y$, or not to have two terms in the product or to drop a 2. Although virtually all candidates differentiated x^2 correctly, the constant in terms of e caused problems for many; they “differentiated” it despite being told it was constant in the question. Most attempted to solve their expression for

$\frac{dy}{dx}$ and of

the many who had incorrectly written $\frac{dy}{dx} =$ in their opening line of implicit differentiation, most ignored it to get a correct expression, and thus it was condoned. Many were unable to obtain a fully correct solution as they still had their constant term, C in terms of e , in their expression.

In part (c) candidates had to have the expression for $\frac{dy}{dx}$ correct from part (b) in order to achieve

both marks. Most substituted the given values of x and y into their expression and knew it should come to zero, and stated it was zero even if it clearly wasn't, rather than looking for an error. About 30% of the candidates gave a fully correct solution to the whole question. A few who otherwise had correct solutions, failed to bring the solution to a conclusion, although most did state a stationary point required the derivative to be zero, or equivalent.

Q9.

In part (a), almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

In part (b), those who substituted $t = 2$ into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt} = 0$ is the

condition for a stationary point. Some **assumed** that a stationary point occurred when $t = 2$, went straight to the test for maximum or minimum and only scored half the marks.

It was advisable to use the second derivative test here; those who considered values of $\frac{dy}{dt}$ on either side of $t = 2$ usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c)(i), the concept of ‘rate of change’ was not understood by many who failed to realise the need to substitute $t = 1$ into $\frac{dy}{dt}$. Some candidates wrongly substituted $t = 1$

into the initial expression for y or into their expression for $\frac{d^2y}{dt^2}$ and these candidates

were unable to score any marks at all on this part. Even those who used $\frac{dy}{dt}$ sometimes made careless arithmetic errors when adding three numbers.

In part (c)(ii), it was not enough to simply write the word “increasing”: some explanation

about $\frac{dy}{dt}$ being positive was also required. Some candidates erroneously found the value of the second derivative when $t = 1$ or calculated the value of y on either side of $t = 1$.

Q10.

In part (a)(i), many candidates did not realise that differentiation was required in order to find the gradient of the curve, but instead erroneously used the coordinates of O and A . Some tried to rearrange the terms but usually made sign errors in doing so. In part (a)(ii), those who had the correct gradient in part (a)(i) were usually successful in finding the correct equation of the normal, though not everyone followed through to the required form, and sign errors were common. However, most obtained at least a method mark here, unless they found the equation of the tangent. The main casualties were once again those who always use the $y = mx + c$ form for the equation of a straight line.

In part (b)(i), most candidates were well drilled in integration and scored full marks, although some wrote down terms with incorrect denominators. The limits 2 and 0 were usually substituted correctly, but it was incredible how many could not evaluate $32 - 38 - 8$ without a calculator. The correct value of the integral was -14 , but far too many thought that they had to change their answer to $+14$ and so lost a mark. A small number of candidates differentiated or substituted into the expression for y rather than the integrated function.

In part (b)(ii), there was still some apparent confusion about area when a region lies below the x -axis. The area of the triangle was 6 units and hence the area of the shaded region was $14 - 6 = 8$ but, not surprisingly, there were all kinds of combinations of positive and negative quantities seen here. It was worrying to see so many candidates failing to calculate the triangle area, with several finding the length of OA instead. Some able candidates found the equation of OA and the area under it by integration, but this was not the expected method.

Q11.

There were many correct answers for the derivative in part (a) although the incorrect answer $-3x^{-2} + 48$ was not a rarity. Candidates who wrote down $y = x^{-3} + 48x$ before differentiating generally scored more marks than those who just stated an answer. Many

candidates appreciated the need to put $\frac{dy}{dx} = 0$ in part (b) but poor algebraic manipulation or a failure to include ' $\pm$ ' frequently resulted in low marks for this part of the question.

Even those who obtained the correct coordinates of the stationary points did not always appreciate that the required tangents had equations $y = \pm 32$. Although the marks for finding the equation of the normal in part (c) were generally better than for the work in part (b), it was surprising to see a significant minority giving the equation of the tangent at the given point.

Q12.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and, if there were errors, it was usually the final accuracy mark that was lost through the omission of brackets at some stage in the solution.

Part (b) was not answered as well as part (a). Although many candidates were successful in factorising the required quadratic, there were also many solutions which were accompanied by terms in e^{-4x} . Where marks were lost, it was mainly due to incorrect signs, though some candidates did manage to obtain follow through marks.

A few candidates attempted to substitute their values of x back into the derivative rather than y . Many candidates stopped when they had found the two values of x , presumably thinking they had finished the question.

Q13.

There were many correct answers to this question, although the amount of working shown varied considerably; some, for example, went into u and v for the product whilst others just wrote all the derivatives down. Those who had the derivatives all correct usually went on to calculate the gradient correctly, although mistakes were seen in handling $e^0 - 1$.

The common error in the differentiation was in the e term where the $\frac{dy}{dx}$ was not attached or it became ye^x or similar. Some candidates differentiated the expression with respect to y , and inverted the result for the gradient, which was both perceptive and, of course, acceptable. Some candidates thought they should take logs of both sides first, and often went on to create nonsense, although those who had recognisable derivatives in an attempt at the chain rule could gain partial credit.

Q14.

In part (a)(i), most candidates stated the formula for $\sin 2\theta$ correctly although not always in its simplest form.

In part (a)(ii), although many candidates knew the formula for $\cos 2\theta$ in at least one its correct forms, several gave $\cos^2 \theta + \sin^2 \theta$, and later in the question just gave the value of

$\cos 2\theta$ as 1 without apparently thinking something could be wrong.

In part (a)(ii), many candidates had problems. It appeared that the 3-4-5 triangle was either not well known or not recognised as relevant; although some candidates did draw this to justify their answer others used $\sin^2 \theta = 1 - \cos^2 \theta$. Many candidates worked backwards from the answer given, to find a value for $\sin \theta$, which gained no credit nor did going via an angle from the given value of $\cos \theta$. Many candidates omitted to find $\cos 2\theta$ or, as above, gave the value as 1, or another incorrect value from an incorrect expression. Those who did find the correct value for $\cos 2\theta$ often dropped the minus sign in later working.

In part (b)(i), the use of the chain rule with these parametric equations was generally done well with most candidates completing it correctly. Some candidates apparently tried to integrate rather than differentiate and some made sign errors when differentiating. A few candidates had the chain rule expression the wrong way up.

In part (b)(ii), although some neat and concise correct answers were seen, many candidates got in a mess trying to use angles and not working in exact fractions. Some confused the value of the sine and cosines involved with the angles writing terms such as $\sin \frac{3}{5}$. Some left a mix of fractions and decimal numbers in their final answer, which is bad practice but was accepted if the values of the coordinates at P were correct.

Q15.

Part (a)(i) Most candidates were able to integrate the expression with only the weakest candidates unable to do this basic integration. Poor notation was used with many including the integral sign after integrating. It would have been thought that this bad habit would have been corrected by the time of the examination. Many candidates did not find the actual value of the definite integral until part (a)(ii) and on this occasion full credit was given. It was alarming that many candidates who had correct fractions were unable to combine these to give a value of 8.4 or equivalent. Weaker candidates were seen

substituting values into the expression for y or $\frac{dy}{dx}$, showing a complete lack of understanding.

Part (a)(ii) It was necessary to consider a rectangle of area 18 and then to subtract their answer from part (a)(i) in order to obtain the area of the shaded region. Many believed that the area of the rectangle was 9 and others failed to do this basic subtraction correctly, even when their answer to part (a)(i) was correct.

Part (b)(i) Many candidates did not realise the need to find $\frac{dy}{dx}$ before substituting the value $x = 1$ and thus failed to score some easy marks for finding the gradient of the curve. A substantial number of candidates tried to calculate the gradient of the straight line between two points on the curve and scored no marks for this.

Part (b)(ii) Unfortunately many candidates tried to find the equation of the normal instead of the tangent to the curve. Otherwise, since there was a generous follow through in this part of the question, most were able to score this final mark. The only exceptions were those who insisted on using $y = mx + c$ where poor arithmetic often prevented them from

finding a value for c .

Q16.

Part (a)(i) Usually after a few abortive attempts many candidates realised that they had to add together the areas of the various faces. Once they had the correct expression for the total surface area most candidates were able to obtain the printed result. There was clearly some fudging on the part of weaker candidates and they could earn little more than a single method mark.

Part (a)(ii) This was surprisingly one of the biggest discriminators on the paper with only the best candidates being able to obtain the correct expression for V . Trying to make y the subject of the equation from part (a)(i) caused problems for many who took this approach; others substituted for xy in the formula $V = 6x(xy)$ and were often more successful. Once again it was quite common to see totally incorrect expressions being miraculously transformed into the printed answer.

Part (b)(i) Most candidates scored full marks for this basic differentiation although it was

not always clearly identified as $\frac{dV}{dx}$ in their working.

Part (b)(ii) Most substituted $x = 2$ into their expression for $\frac{dV}{dx}$ and found the value to be zero. It was then necessary to make a statement about the implication of there being a stationary value in order to score full marks.

Part (c) Some made a sign error when finding the second derivative, but the majority of candidates scored full marks in this part. Credit was given if the correct conclusion was drawn from the sign of their second derivative, provided no further arithmetic errors occurred.

Q17.

Most candidates coped with the negative fractional powers well, differentiated correctly and found an equation of a normal, although there were a few equations of tangents presented as final answers to part (b)(ii). A common arithmetical slip in part (b)(ii) was ‘

$\frac{1 + \sqrt{1}}{1}$ ’, incorrectly evaluated as 1. The expression for the second derivative was often correct in part (c)(i).

Part (c)(ii) was discriminating. The most popular approaches were to either put $x = 1$ in

$\frac{d^2y}{dx^2}$, or to equate $\frac{d^2y}{dx^2}$ to zero and solve, for which no marks were scored.

A significant number of candidates did, however, earn some credit by making a more general comment on the sign of the second derivative but many lacked rigour and, in particular, assumed the second derivative has to be negative for a maximum, overlooking the possibility that the second derivative may be zero at a maximum.

Q18.

Many candidates lost marks on this question from careless work when handling the number of e^{2x} terms involved.

- (a) Most candidates obtained the correct coordinates although $(\ln x, 0)$ was a common error.
- (b) This was generally well answered by those candidates able to use the quotient rule. Some errors with the numerator being reversed to $(\ln x - 1)$ meant candidates lost accuracy marks later in the question. Many candidates obtained 3 marks for reaching $\ln x - 1 = 0$ and many went on to find $x = e$. However many stopped here without finding the value of y .
- (c) Although many correct solutions were seen some candidates lost marks through leaving $(e^{3x})^2$ rather than simplifying it. There was also a large number of candidates who did not answer part (b) but started again here, used the quotient rule correctly and went on to then answer the remainder of the question correctly. They seemed to realise that they needed the derivative to find a gradient but not for finding stationary points

Q19.

- (a) The vast majority of candidates completed this question successfully and as expected, by finding the two derivatives with respect to t and combining them using the chain rule. The more common errors were to omit the squared powered from $6t^2$ or the negative sign in -3 . Some candidates lost a mark through not simplifying their final result. Relatively few candidates made the error of multiplying the derivatives or having the chain rule upside down. Very few candidates took the approach of eliminating t and seeking an equation $y = f(x)$. Of those who did, hardly any were successful in differentiating their expression.
- (b) The vast majority of candidates was successful here too, substituting $t = 1$ immediately and obtaining the gradient of the normal correctly and then a correct form of the equation of the normal. Those few candidates who made no attempt to find the gradient of the normal and just used the tangent gained no credit. The fairly rare error in finding the gradient of the normal was to omit the negative sign. Some candidates also failed to substitute for the given value for t and gained no credit.
- (c) Various correct approaches were seen with the most efficient being to obtain t in terms of x and substitute into $y = f(t)$, and many candidates did this correctly. The common errors were to omit or have an unclearly positioned negative sign, or to make an error in the power. Other attempts at elimination were generally not so successful. Many candidates wasted time through proceeding to expand their cartesian equation, or just generally and unnecessarily manipulating it algebraically. Once t is eliminated, the resulting equation is a cartesian equation.

Q20.

- (a) There was a mixed response to this question with most candidates able to solve for x correctly algebraically but with their calculator in degrees mode. Some thought the cube root could be positive or negative, and some found the square root. Many left their answer as $\sqrt[3]{7 - \cos \pi}$ presumably regarding this as exact, rather than

evaluating to the value 2.

- (b) The vast majority of candidates attempted to differentiate the expression as given. Those few who manipulated the equation and solved it for x or y , and then attempted differentiation usually made little successful progress, being unable to cope with the quotient rule and chain rule involved. Of those candidates who took the expected approach, most could differentiate $x^3 y$ correctly, the common errors being $\frac{dy}{dx}$ attached to the wrong term, derivatives multiplied rather than added, or an incorrect power of x . Many candidates couldn't differentiate the $\cos(\pi y)$ term correctly, apparently often not realising that π is a number when using the chain rule, some turning π into $\frac{1}{\pi}$ and some just making a sign error. Most knew that the 7 differentiated to 0, although most didn't make this explicit, it was just implied in their solution for $\frac{dy}{dx}$. Most attempted to solve for $\frac{dy}{dx}$ before substituting values for x and y , and although unnecessary, gained credit as long as the 7 was no longer present.

Q21.

In part (a) almost all candidates were able to find the first and second derivative correctly, although there was an occasional arithmetic slip and some could not cope with the fraction term.

Those who substituted $t = 3$ into $\frac{dx}{dt}$ in part (b) did not always explain that $\frac{dx}{dt} = 0$ is the condition for a stationary point. Some assumed that a stationary point occurred when $t = 3$ and went straight to the test for maximum or minimum and only scored half the marks. It was advisable to use the second derivative test here; those who considered values of $\frac{dx}{dt}$ on either side of $t = 3$ usually reached an incorrect conclusion because of the proximity of another stationary point.

In part (c) the concept of "rate of change" was not understood by many. Approximately equal numbers of candidates substituted $t = 1$ into the expression for $\frac{dx}{dt}$ or $\frac{d^2x}{dt^2}$ and so only about half of the candidates were able to score any marks on this part. Those who used $\frac{dx}{dt}$ often made careless arithmetic errors when adding three numbers.

In part (d), as in part (c), candidates did not realise which expression to use and many wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a particular point.

Q22.

In part (a)(i) the majority of candidates realised the need to find the value of $f(x)$ when $x =$

2. However, it was also necessary, after showing that $f(2) = 0$, to write a statement that the zero value implied that $x - 2$ was a factor. It was good to see more candidates being aware of this.

In part (a)(ii), those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(i) a surprising number of candidates failed to see the need to differentiate in order to find the gradient at Q . Those who attempted to find $\frac{dy}{dx}$ sometimes wrote it as $3x^2 + x$, but usually were aware of the need to substitute $x = 2$.

In part (b)(ii) those who had the correct gradient in part (b)(i) were usually successful in finding the correct equation of the tangent, and most obtained at least a method mark here.

In part (b)(iii) most were well drilled in integration and earned full marks, although some wrote $\frac{x^4}{4} + \frac{x^2}{2} - 10$ and others gave $\frac{x^4}{4} + \frac{x^2}{2} - \frac{10^2}{2}$ as their answer. For part (b)(iv) the correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 0 and 2 and it was incredible how many could not evaluate $6 - 20$ without a calculator. Very few candidates realised the need to show clearly that, although the integral from 0 to 2 gave a value of -14 , the area of the shaded region was 14. A separate statement was needed and those who simply wrote $4 + 2 - 20 = -14 = 14$ did not score full marks. Those more able candidates who made a statement about the region being entirely below the x -axis and who subsequently evaluated the integral from 2 to 0 correctly scored full marks.

Q23.

This long calculus question with a final part covering transformations was a relatively high source of marks for many candidates. Most candidates were able to correctly differentiate $2x^{\frac{3}{2}}$, but not all evaluated it correctly at $x = 4$. The majority of candidates showed a good understanding of the method involved in finding the equation of the normal in part (a)(ii), although some others did not use the correct gradient. Although the vast majority of

candidates knew the rules for integrating $8x^{\frac{1}{2}}$, there was a significant number of such candidates who did not simplify the coefficient correctly.

It was not uncommon to see either ' $\frac{8}{3}x^{\frac{3}{2}}$ ' not simplified or simplified incorrectly to ' $12x^{\frac{3}{2}}$ '.

Part (b)(ii) was generally answered better than expected, with a majority of candidates applying a correct method for finding the area of the shaded region. Although many candidates realised that the transformation was a 'translation', there were frequently sign errors in the vector, or components were interchanged. It was also disappointing to see a significant increase in the number of descriptions that involved more than one transformation which, as usual, was given no marks; the two transformations quoted were translation and a stretch.

Q24.

There was a significant improvement from June 2008 in candidates' solutions to the binomial expansion topic. Part (a) was well answered, although a lack of brackets continues to be the main source of errors. Many candidates illustrated that they knew how to use their answer to part (a) to write down the expansion of $(1 - 2x)^4$, but not all solutions which finished with the printed answer were entirely convincing.

Part (c) was not always attempted and, even when attempted, the majority failed to convincingly show that there was only one stationary point. Many solutions included the correct equation ' $96x + 128x^3 = 0$ ', but a significant proportion of these lost the ' $x = 0$ ' root

and, by erroneously changing a sign, ended up with a value for x of the form $\frac{\sqrt{3}}{2}$.

Q25.

In part (a), most candidates used the product rule correctly, but at this stage $e^{2x} \neq 0$ was rarely seen, although many candidates obtained the correct quadratic equation. Factorisation, or use of the formula, was usually done well and many candidates obtained both of the correct solutions. Where candidates seemed unsure of what to do with the e^{2x} , many expressions involving the use of $\ln e^{2x} = 2x$ were seen together with the formation of a cubic expression in x . A third value of $x = 0$ was also seen by candidates who thought this was the solution of $e^{2x} = 0$.

Part (b) was generally well answered by those candidates able to do the product rule in part (a). Some errors with the coefficients meant candidates lost accuracy marks.

In part (c), although many correct solutions were seen some candidates lost marks

through incorrect evaluation of $\frac{d^2 y}{dx^2}$.

Q26.

The implicit differentiation in part (a) was done well, with many candidates deriving the derivative as a function of x and y correctly, although this was not necessary to find its

value. The common error was to have $\frac{d}{dx}(x^2 y) = 2xy \frac{dy}{dx}$, or a similar error in an attempt at a two term product rule. The derivatives of $3y^2$ and $2x + 1$ were usually correct, although a few candidates neglected to differentiate the right hand side.

Very few candidates brought a spurious $\frac{dy}{dx}$ into their equation, although many did start

their answer with " $\frac{dy}{dx} =$ " before proceeding correctly. A few candidates did attempt to rearrange the given equation to the form $y = f(x, y)$ and differentiate using the quotient rule, but this was usually incorrect, particularly in the handling of the y terms.

The equation was given in the form considered to be of most help to candidates, who should be advised not to attempt to rearrange such equations. Relatively few candidates could successfully complete part (b), although most did equate their expression for the

derivative to zero. Some then cross multiplied as if 0 were 1, and many others just abandoned their solution, not realising that it was the numerator that was zero. Those who did get to a correct form of $xy = 1$ usually completed successfully, although some tried substituting into the derivative rather than the original equation, and some seemed to be looking for an expression in y , rather than x as required.

Q27.

In part (a) almost everyone obtained the correct expression for $\frac{dy}{dx}$, although a few spoiled their solution by dividing each term by 5 or adding “+ c ” to their answer.

In part (b) most candidates substituted $x = -2$ into their expression for $\frac{dy}{dx}$, but, in order to score full marks, it was necessary to show $(-2)^4$ written as 16 or to show that $\frac{dy}{dx} = 80 - 80 = 0$ and then to write an appropriate conclusion about P being a stationary point.

For part (c) many candidates simply wrote down an expression for $\frac{d^2y}{dx^2}$ in terms of x when answering part (i) and only evaluated the second derivative when determining the nature of the stationary point in part (ii). On this occasion full credit was given, but candidates need to realise what is meant by the demand to “find the value of” since this may be penalized in future examinations.

In part (d) some candidates failed to find the y -coordinate of P , which was necessary in order to find the equation of the tangent. It was pleasing to see most candidates using the value of $\frac{dy}{dx}$ when $x = 1$, but unfortunately many tried to find the equation of the normal instead of the tangent to the curve.

Q28.

In part (a) most candidates differentiated the equation of the curve correctly and the majority of these recognised that at the maximum point the value of $\frac{dy}{dx}$ was 0, although some worked with $\frac{d^2y}{dx^2} = 0$.

In part (b) many candidates had difficulty solving the equation $22.5x^{\frac{1}{2}} - 2.5x^{\frac{3}{2}} = 0$. There was some invalid squaring of expressions and terms and also it was not uncommon to see $x^{\frac{3}{2}} \div x^{\frac{1}{2}}$ simplified incorrectly either to x^3 or to x^2 .

With the equation of the tangent printed in part (d) candidates had to convince examiners

that they were finding the value of $\frac{dy}{dx}$ at $x = 1$ for the gradient of the tangent in part (c). Although not all candidates did this convincingly, the majority did show sufficient evidence to reach the printed equation.

The final part of the question proved to be a good discriminator at the top grade level. Although some excellent solutions were seen these were rare. The most common wrong method was to assume that the length of RM was either equal to the length of PM or equal to half the length of PM . Unfortunately a minority of candidates who produced better solutions which started with the correct equation ' $162 = 20x - 6$ ' failed to score the final mark due to an inability to solve this equation correctly.

Q29.

Part (a)(i) was not very well answered by the majority of candidates. Fully correct responses were seen but it was usually from candidates who successfully rearranged the equation into the form $f(x) = 0$, which was seen in a number of acceptable forms. Where candidates simply substituted in the two values given into the LHS they obtained 1 and 0 but still indicated that this was a change of sign and therefore there was a root. Most candidates achieved at least one mark but lost the second by simply stating change of sign therefore a root.

Very few incorrect responses were seen in part (a)(ii).

Part (a)(iii) was very well answered. The main error was with the candidates who used degrees rather than radians

Part (b)(i) was very well answered, with most candidates successfully using the quotient rule. Where errors occurred it was usually through missing brackets, although many candidates were able to recover at some stage in the working.

As part (b)(ii) followed on from part (i) most candidates were able to obtain the mark for substitution; even those who had incorrectly simplified their work. Many fully correct responses were seen. Several candidates also stopped after the substitution of $x = 0$ and left their answer as -2 .

Q30.

In part (a) almost all candidates obtained $dx/dt = -1/t^2$. However, most candidates could not handle the fraction and indices correctly when finding dy/dt . The most common mistake was to state $dy/dt = 1 - 2/t^2$. Sometimes this "became" $1 - 1/2t^2$ but, by then, the mark had been lost. The t^2 sometimes migrated on to the numerator. Almost all candidates knew they needed to divide dy/dt by dx/dt to get dy/dx . Full marks were awarded for an unsimplified form of dy/dx .

A few candidates confused differentiation and integration giving the "derivatives" in terms of $\ln t$. Some candidates proceeded by first attempting to find the cartesian equation, which was valid if the result was given in terms of t . Surprisingly many of these candidates then didn't make much progress in part (c). In part (b) the majority of candidates found the equation of the normal, and not the tangent, and the relationship between the gradients was well known.

Very few found an incorrect y value at $t = 1$. Most errors were due to an incorrect dy/dx from part (a), but some credit was given for method and 3 marks were still available. These were earned by most. Some candidates had an expression for dy/dt in which the t 's cancelled, yet it didn't apparently occur to them that an error had been made. Most candidates began by writing the cartesian equation in an unsimplified form in part (c). Many were defeated by the fraction in the denominator and so could not cope with $1/(2(1/x))$. Errors were often made when attempting to multiply through by $2x$. Some candidates chose to substitute for t in the given equation, but not all could deal with the algebra involved; $k = 1$ or $k = -2$ were common wrong conclusions. Pleasingly most candidates who did attempt this question made an algebraic approach and very few just substituted $x = 1$ and $y = 3/2$ to find the value of k , which gained no credit.

Q31.

It was pleasing to see complete correct solutions from a large number of candidates. Clearly implicit differentiation of equations such as this is a topic that had been learnt well. It was a pity that many candidates wasted time in rearranging the equation in x , y and dy/dx to isolate dy/dx . A quicker method was to substitute $x = 1$ and $y = 3$ immediately after differentiating. A few candidates simply wrote $8xdx + 2ydy = 3ydx + 3xdy$ or similar, which is incorrect notation and scored no marks.

The common errors were to leave the 4 in the differentiated expression, or not to differentiate the product term correctly, attaching dy/dx to the wrong term or having just one term. The few candidates who chose to rearrange the given equation and solve it for y before differentiating generally made little further progress, although such methods would be valid if carried out correctly.

Q32.

In part (a), most candidates were able to find the correct expression for $\frac{dy}{dx}$, although there were some who left $+ 5$ in their answer or added $+C$.

In part (b), It had been expected that candidates would solve the equation $\frac{dy}{dx} = 0$ and obtain the equation $x^3 = 8$ and hence deduce that $x = 2$. It seemed, however, that many were unable to formulate an appropriate equation, but merely spotted the correct answer: $x = 2$. This was not penalised on this occasion, provided that the candidate stated clearly that the x -coordinate of M was equal to 2.

In part (c)(i), the expression for $\frac{d^2y}{dx^2}$ was usually correct.

In part (c)(ii), although the method was left open, most candidates found the value of the second derivative when $x = 2$ and correctly concluded that M was a minimum point.

In part (c)(iii), some candidates were not aware of the need to find the value of $\frac{dy}{dx}$ when

$x = 0$ in order to ascertain whether the curve was increasing or decreasing at that point.

Q33.

In part (a)(i), a few candidates ignored the request to use the factor theorem and scored no marks for using long division. It was necessary to make a statement that “ $x + 1$ is a factor”, after showing that $f(-1) = 0$, in order to score full marks.

Part (a)(ii) was not answered as well as similar questions in previous years. Perhaps the sketch lured some into trying to write down three factors without any further working, rather than using the intermediate step of showing that $p(x) = (x + 1)(x^2 - x - 6)$ before writing $p(x)$ as a product of three factors. Many who tried long division were flummoxed by there being no x^2 term.

In part (b)(i), those who had the correct linear factors in part (a)(ii) usually wrote down correctly that A had coordinates $(-2, 0)$, although some carelessly wrote the point as $(0, -2)$.

Many candidates simply found an indefinite integral in part (b)(ii) and then a definite integral in part (b)(iii). The two parts were generously treated holistically when candidates did this. The fractions once again caused problems to most candidates who are so used to having a calculator to do this work for them. It was very rare to see the correct answer of -32 for the definite integral.

In part (b)(iii), many lost out on an easy mark because they rolled their two sections into one: those who wrote “integral = $-32 = 32$ ” gained full credit for part (b)(ii) but did not score the mark in part (b)(iii). It was necessary to give a positive value for the area of the region and to make this explicit. In anticipation of a lot of wrong answers in part (b)(ii), a follow through mark was awarded in part (b)(iii): for example, if a candidate’s answer in part (b)(ii) was -20 and they concluded that the area was 20 in part (b)(iii), they scored the mark.

In part (b)(iv), most candidates differentiated correctly, but quite a few thought that $3(-1)^2 - 7$ was equal to -10 and thus obtained the wrong gradient of the curve.

In part (b)(v), a large number of candidates found the correct equation of the normal but some still confused tangents and normals and consequently thought that the gradient of the normal was equal to -4 . It was quite common for weaker candidates to either negate their gradient or take the reciprocal but to fail to do both.

Q34.

Generally, most candidates scored relatively high marks for this long structured calculus question. Parts (a)(i), (ii) and (iii) were answered very well, although some used the gradient of the normal in part (a)(iii) as -2 and so in part (a)(iv) obtained a positive value for the y -coordinate of Q . A glance at the diagram should have indicated that such a value was incorrect.

Part (a)(v), which required candidates to find the x -coordinate of the maximum point, M , was generally not answered with any confidence. To score any marks for this part, candidates were required to go beyond just equating their answer for part (a)(i) to zero. Many candidates again incorrectly believed that for a maximum point the second derivative was zero. Candidates generally found the correct answer to the indefinite

integral in part (b)(i), but a significant minority forgot to add the area of the triangle to their 'area under the curve' in part (b)(ii) of the question.

Q35.

Most candidates were correct in part (a)(i), with very few arithmetic errors seen. Similarly, the differentiation required for part (a)(ii) was usually done correctly, with the common

error being $\frac{d}{dt}\left(\frac{1}{t^2}\right) = -\frac{1}{t}$. Most candidates used the chain rule correctly, although some

poor algebra was seen in inverting $\frac{dx}{dt}$, with $\frac{1}{2-2t^{-3}} = \frac{1}{2} - \frac{1}{2t^{-3}}$ being seen quite often; if candidates had not made it clear that this is what they believed the reciprocal to be, they were denied the chain rule method mark. However, most candidates scored the mark for finding a tangent, although a few found an equation for the normal.

Those who chose to find c in $y = mx + c$ instead of using $(y + 3) = m(x - 5)$ were prone to error. Part (b) was much less successful for most candidates. Despite the required form of the answer being given, few candidates chose to find $(x + y)$ and $(x - y)$ in terms of t and apply the result to the given equation. Of those that did do this, there were many errors in the squaring and multiplication. Some candidates tried to find t in terms of x and y , often getting lost algebraically in their attempt and abandoning it; some did manage to complete this method successfully, but it was time consuming. Others simply did not attempt this part of the question.